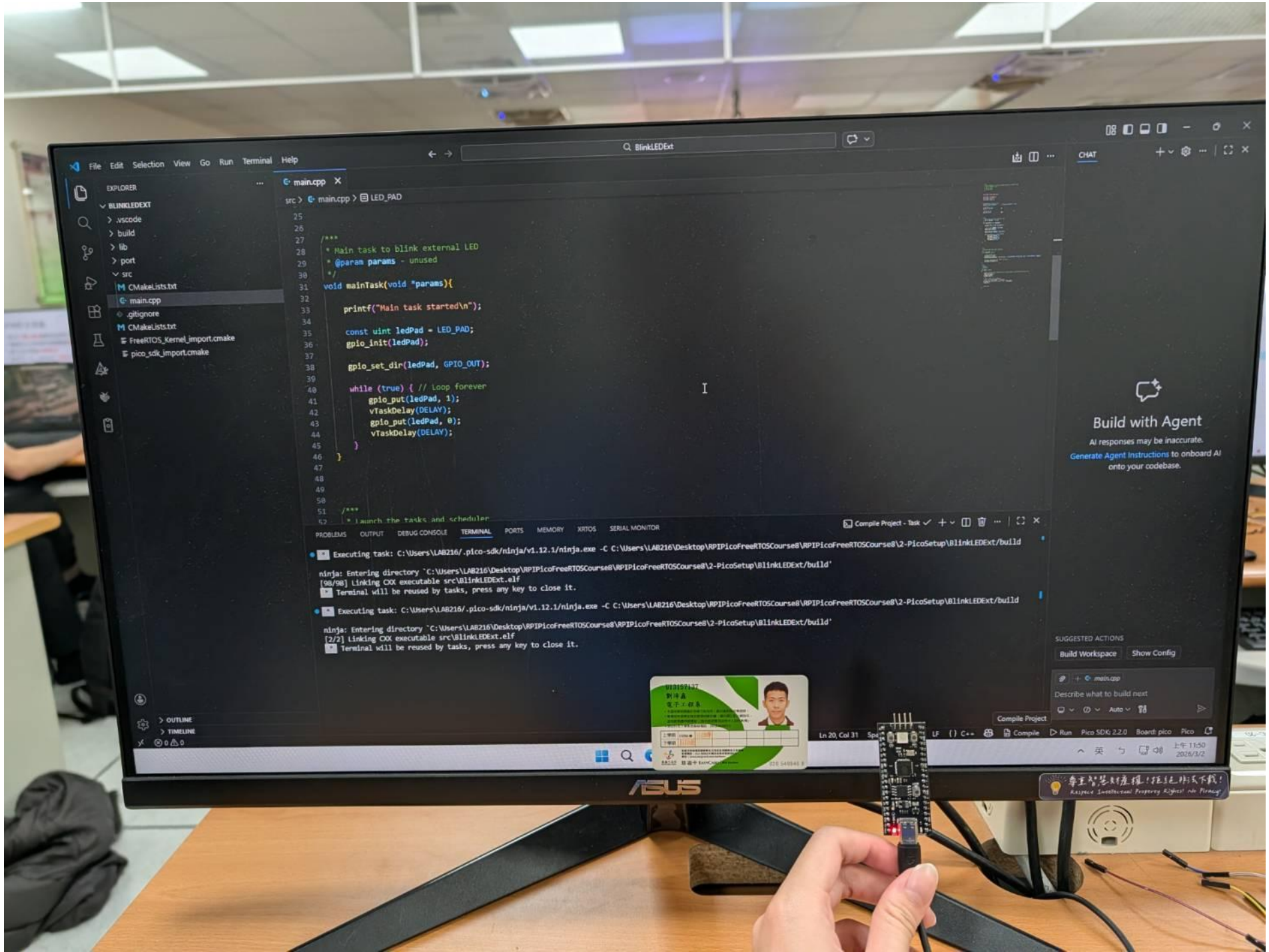
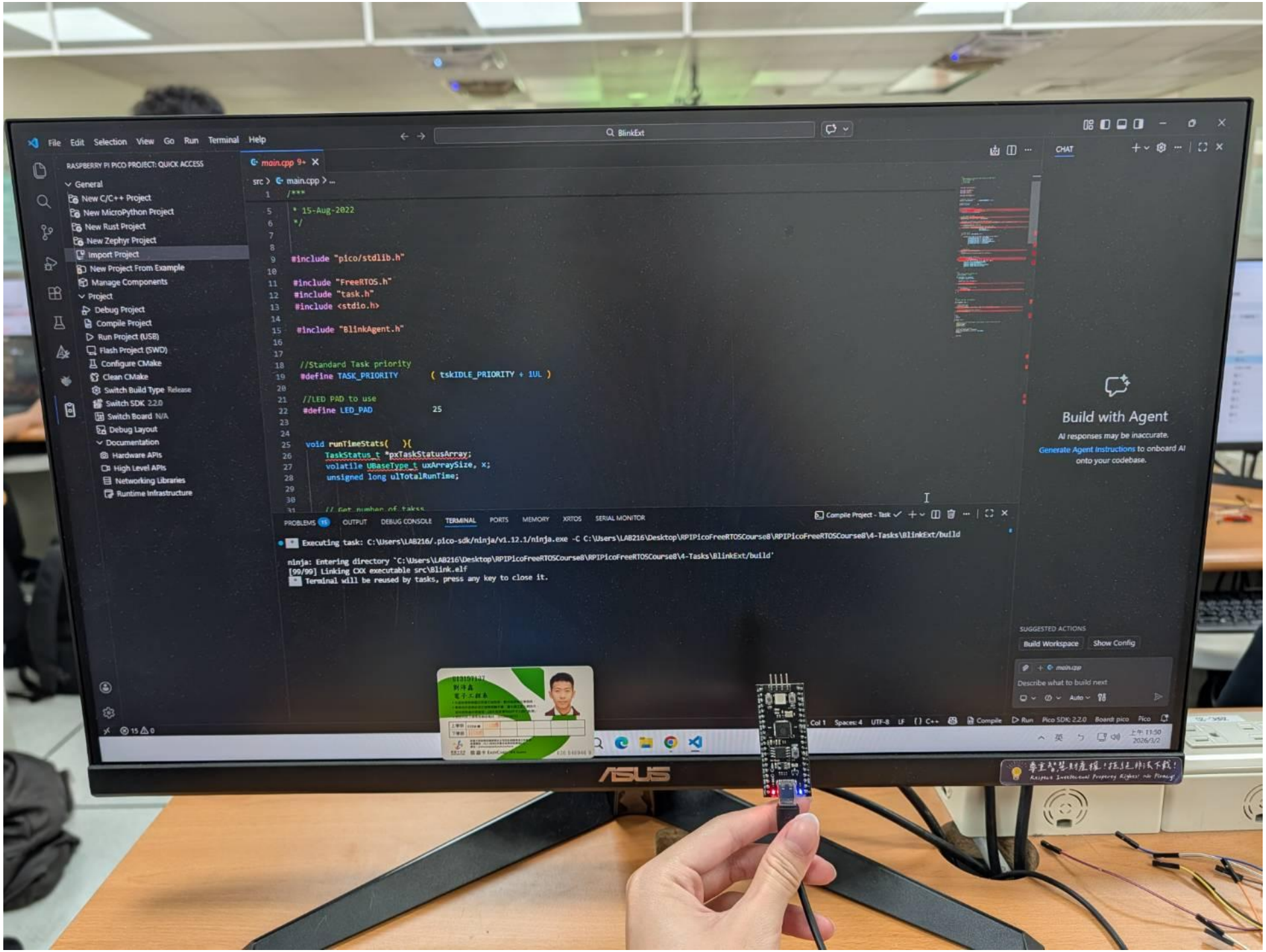
OS-3 比較題

U13157137 劉沛鑫


File Edit Selection View Go Run Terminal Help
BlinkLEDExt
EXPLORER
BLINKLEDEXT
.vscode
build
lib
port
src
CMakeLists.txt
main.cpp
.gitignore
CMakeLists.txt
FreeRTOS_Kernel_import.cmake
pico_sdk_import.cmake
src > main.cpp > LED_PAD
/***
* Main task to blink external LED
* @param params - unused
*/
void mainTask(void *params){
printf("Main task started\n");
const uint ledPad = LED_PAD;
gpio_init(ledPad);
gpio_set_dir(ledPad, GPIO_OUT);
while (true) { // Loop forever
gpio_put(ledPad, 1);
vTaskDelay(DELAY);
gpio_put(ledPad, 0);
vTaskDelay(DELAY);
}
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
Compile Project - Task
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build'
[98/98] Linking CXX executable src\BlinkLEDExt.elf
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build'
[2/2] Linking CXX executable src\BlinkLEDExt.elf
Terminal will be reused by tasks, press any key to close it.
CHAT
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
SUGGESTED ACTIONS
Build Workspace
Show Config
Describe what to build next
Compile Project
OUTLINE
TIMELINE
Ln 20, Col 31
LF
Compile
Run
Pico SDK: 2.2.0
Board: pico
ASUS


File Edit Selection View Go Run Terminal Help
BlinkExt
RASPBERRY PI PICO PROJECT: QUICK ACCESS
General
New C/C++ Project
New MicroPython Project
New Rust Project
New Zephyr Project
Import Project
New Project From Example
Manage Components
Project
Debug Project
Compile Project
Run Project (USB)
Flash Project (SWD)
Configure CMake
Clean CMake
Switch Build Type Release
Switch SDK 2.2.0
Switch Board N/A
Debug Layout
Documentation
Hardware APIs
High Level APIs
Networking Libraries
Runtime Infrastructure
main.cpp 9+
src > main.cpp > ...
/***
* 15-Aug-2022
*/
#include "pico/stdlib.h"
#include "FreeRTOS.h"
#include "task.h"
#include <stdio.h>
#include "BlinkAgent.h"
//Standard Task priority
#define TASK_PRIORITY (tskIDLE_PRIORITY + 1UL)
//LED PAD to use
#define LED_PAD 25
void runTimeStats( ){
TaskStatus_t *pxTaskStatusArray;
volatile UBaseType_t uxArraySize, x;
unsigned long ulTotalRunTime;
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
Compile Project - Task
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\4-Tasks\BlinkExt/build
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\4-Tasks\BlinkExt/build'
[99/99] Linking CXX executable src\Blink.elf
Terminal will be reused by tasks, press any key to close it.
CHAT
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
SUGGESTED ACTIONS
Build Workspace
Show Config
Describe what to build next
ASUS

README

實習報告：Pico LED Blink 專案開發與架構比較

學生姓名： 劉沛鑫 **學號：** U13157137

1. 實驗目標

透過 Raspberry Pi Pico 控制 GPIO 25 (內建藍色 LED)，對比基礎 SDK、模組化開發與 FreeRTOS 即時作業系統的差異，並熟悉硬體燒錄 SOP（BOOT + RST 模式）。

---

2. 三種開發方案架構比較表

比較維度	(Ch0) SDK Example	(Ch2) BlinkLEDExt	(Ch4) Blink (FreeRTOS)
技術層次	原生 C-SDK 範例	外部引腳模組化	**RTOS 即時作業系統**
程式邏輯	無限 while(1) 迴圈	結構化定義與封裝	**任務制 (Task-based)**
時間管理	sleep_ms (阻塞 CPU)	sleep_ms (阻塞 CPU)	vTaskDelay (釋放 CPU)
系統特性	裸機執行 (Single-thread)	裸機執行 (Modular)	**多工處理 (Multi-tasking)**
優點	程式碼極小、反應直接	代碼整潔、易於維護	高擴充性、適合複雜專案

---

3. 操作 SOP 與心得總結

- **硬體進入傳輸模式 (RPI-RP2)**：
  1. 按住 **BOOT** 鍵不放。
  2. 點擊 **RST** 鍵一下。
  3. 放開 **BOOT** 鍵（電腦出現 USB 磁碟）。
- **編譯與燒錄**：在 VSCode 使用 CMake Tools 編譯出 .uf2 檔，直接拖入磁碟即可運行。
- **架構心得**：
  - **Ch0/Ch2 (裸機模式)**：適合簡單控制，但當程式變大時，CPU 會因為等待延遲（Sleep）而無法處理其他事，效率較低。
  - **Ch4 (FreeRTOS)**：這是專業開發的轉折點。透過「任務排程」，LED 閃爍不再佔用全部 CPU 資源，系統可以在閃燈的空檔處理傳感器或網路數據。

---

4. 實作截圖 (請在此處貼上你的 3 張截圖)

[截圖 1：VSCode 編譯成功畫面] > (需拍到 Terminal 顯示 Built target ... 與 [100%])

[截圖 2：檔案總管 UF2 檔案] > (顯示生成出的 Blink.uf2 或 BlinkLEDExt.uf2)

[截圖 3：硬體運行狀態] > (Pico 實體閃燈照片或 VSCode 專案目錄結構)

製作 PDF 小撇步 (符合最小檔案要求)：

1. **調整圖片**：將截圖貼進 Word 後，點擊圖片 -> 圖片格式 -> **壓縮圖片**（選擇 96ppi 或 150ppi）。
2. **另存 PDF**：點選 檔案 > 另存新檔 > 格式選 PDF > 點選 **「最小檔案大小 (線上發佈)」**。